

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
10	(a)			electron promoted to a higher energy level (1)						
				falls back down to a lower energy level (1)	2			2		
	(b)	(i)		each series corresponds to electrons falling to different energy levels	1			1		
		(ii)		energy difference between the shells decreases / energy levels get closer together	1			1		
	(c)	(i)		ionisation of the atom / point at which electron leaves the atom	1			1		
		(ii)		$\Delta E = hf$ and $f = c/\lambda$ (1) $\frac{6.63 \times 10^{-34} \times 3.00 \times 10^8}{92.1 \times 10^{-9}} \quad (1)$ $2.18 \times 10^{-18} \text{ J} \quad (1)$ $1312 \text{ kJ mol}^{-1} \quad (1)$	1					
						3		4	4	